

Sprint 3 Plan

Product Name: Spatial Layout Engine

Team Name: Vision Studio

Sprint Completion Date: May 19, 2026

Revision Number: 1.0

Revision Date: Apr 28, 2026

Goal: Sprint 3 focuses on helping users bring their room vision to life by choosing, placing, and refining furniture within their real space. This sprint emphasizes user confidence, creativity, and practical decision-making by making sure furniture layouts are visually useful, physically realistic, and easier to explore before making real-world purchases.

Tasks

1. As a user, I want to browse furniture options so that I can choose pieces that match my room.
 - 1.1. Create a furniture catalog panel inside the editor workspace. (2 hours)
 - 1.2. Add starter furniture categories such as sofa, bed, table, chair, desk, and storage. (1.5 hours)
 - 1.3. 1.3. Add furniture cards with name, category, dimensions, and placeholder image/model preview. (2 hours)
 - 1.4. 1.4. Connect catalog data to the editor using the furniture metadata schema. (2 hours)
 - 1.5. 1.5. Test catalog browsing, filtering, and item selection flow. (1 hour)Total for Furniture Browsing: 8.5 hours

2. As a user, I want to drag and place furniture into my room so that I can test layouts.
 - 2.1. Build drag-and-drop furniture placement from the catalog into the room editor. (3 hours)
 - 2.2. Add basic furniture movement controls inside the room. (2 hours)
 - 2.3. Add rotation controls for placed furniture. (2 hours)
 - 2.4. Add resize/custom-dimension controls for simple furniture blocks. (2 hours)
 - 2.5. Save the placed furniture position, size, and rotation in the project layout state. (2 hours)
 - 2.6. Test placement behavior across different room shapes and screen sizes. (1.5 hours)Total for Furniture Placement: 12.5 hours

3. As a user, I want furniture fit and collision checks so that I avoid costly mistakes.

- 3.1. Create boundary-checking rules so furniture cannot be placed outside the room. (2.5 hours)
 - 3.2. Create collision detection between furniture items. (3 hours)
 - 3.3. Add visual warnings when furniture overlaps or does not fit. (2 hours)
 - 3.4. Add clear user-facing error messages such as “This item overlaps with another piece” or “This item is too large for the room.” (1.5 hours)
 - 3.5. Test fit validation with different furniture sizes and room dimensions. (2 hours)
- Total for Fit and Collision Checks: 11 hours

- 4. As a user, I want recommendations so that decorating feels easier.
 - 4.1. Create recommendation rules based on room size, available space, and furniture category. (2.5 hours)
 - 4.2. Add a “Recommended for this room” section in the furniture panel. (2 hours)
 - 4.3. Connect recommendations to the user’s room dimensions and existing placed items. (2.5 hours)
 - 4.4. Add simple style tags such as modern, minimal, cozy, neutral, or compact. (1.5 hours)
 - 4.5. Test whether recommendations are physically realistic and user-friendly. (1.5 hours)
 - 4.6. Add interior design elements, including changing the color of walls, wallpaper, wall art, and making sure the general layout of the rooms matches what the user wants (ie feng shui) (2 hours)
- Total for Furniture Recommendations: 12 hours

- 5. As a user, I want auto-generated room layouts so that I can quickly explore ideas.
 - 5.1. Build a basic auto-layout button in the editor. (1.5 hours)
 - 5.2. Create a first-pass layout generator that places selected furniture using room constraints. (3.5 hours)
 - 5.3. Add logic to avoid wall overflow and furniture collisions during auto-placement. (3 hours)
 - 5.4. Allow users to accept, reject, or manually adjust the generated layout. (2 hours)
 - 5.5. Test generated layouts with at least two room types, such as bedroom and living room. (2 hours)
- Total for Auto-Generated Layouts: 12 hours

Total Sprint Estimated Time: 56 hours

Team Roles

Ashley Kim: Product Owner, Frontend Developer, UX/UI Lead, Deliverables Manager

William Liu: Full Stack Developer, Coding Lead, Furniture Placement Lead

Sriya Katreddi: Backend Developer, Data/Validation Developer, Recommendation Logic Support
Ethan Cao: Scrum Master, Backend Developer, Security Lead, System Organizer

Task Assignment

Ashley Kim: User Story 1, Create a furniture catalog panel inside the editor workspace

William Liu: User Story 2, Build drag-and-drop furniture placement from the catalog into the room editor

Sriya Katreddi: User Story 4, Create recommendation rules based on room size, available space, and furniture category

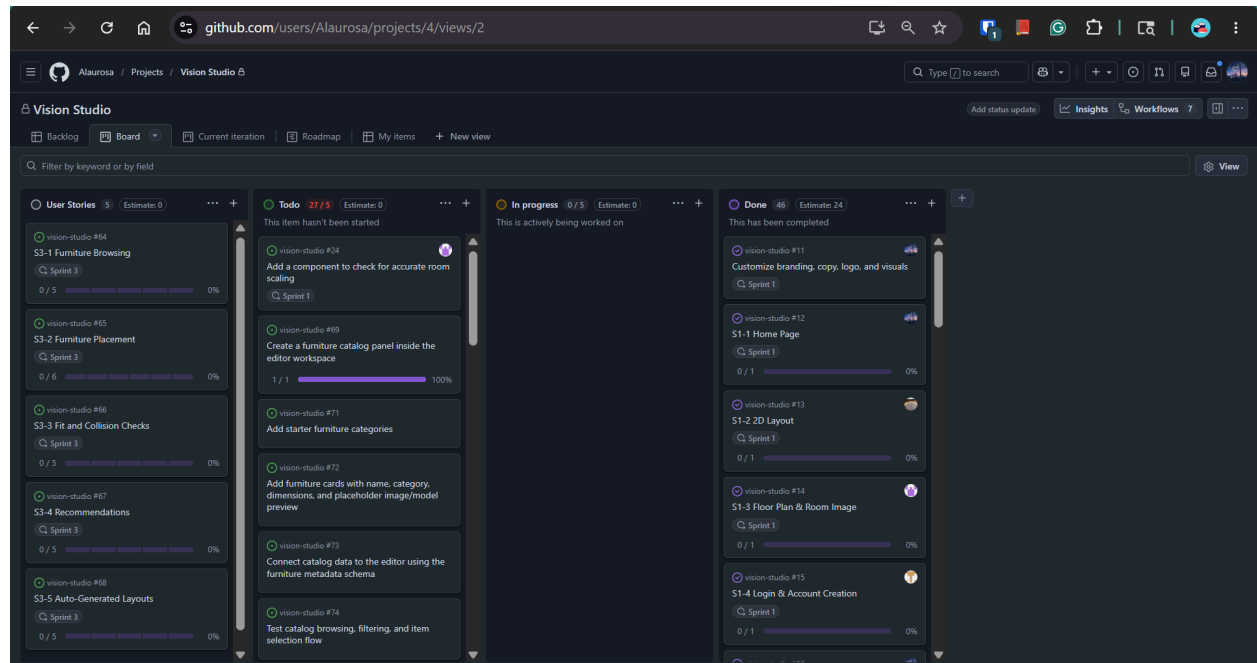
Ethan Cao: User Story 3, Create boundary-checking rules so furniture cannot be placed outside the room

Backlog / Flex Assignment: Whoever finishes first will begin User Story 5, Build the basic auto-layout button and first-pass generated layout flow.

Burnup Chart



Scrum Board



Scrum Times

Scrum Meeting 1: Monday 4:00 PM - 5:00 PM

TA Meeting: Tuesday 11:00 PM - 12:00 PM

Scrum Meeting 2: Tuesday 12:00 PM - 1:00 PM

Scrum Meeting 3: Thursday 1:30 PM - 2:30 PM